

國立台北科技大學

資訊工程系

數位邏輯設計實習報告

Lab 1: Characteristics of Logic Gates

第13組：

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Objective

- To understand the DC & AC characteristics of TTL and COMS gate
- To know how to measure TTL and CMOS parameters with an oscilloscope

Background knowledge

1. Study the definition of and
2. Find the IIL, IIH, VOL, and VOH of 7404 gates in the datasheet
3. Clearly explain the differences between power, ground, input, and output of an 74LS04 gate

Experiments, observations, and discussion

傳遞延遲：

透過將多級反向器串聯，利用示波器測量輸入與輸出波形之間的時間差，藉此計算單個邏輯閘的平均傳遞延遲 t_p 。在環形振盪器實驗中，邏輯閘在不需外部訊號輸入的情況下產生自激振盪。

電壓對延遲的影響：

針對 40 系列 CMOS 邏輯閘，分別在電源電壓 VDD 為 +5V、+10V 及 +15V 的條件下測量延遲。

電壓轉換特性：

利用示波器的 X-Y 模式 觀察 74 系列邏輯閘的電壓轉換曲線，藉此找出邏輯狀態切換的臨界電壓。

環形振盪器的邏輯原理：

環形振盪器必須使用奇數個反向器才能維持振盪。

平均傳遞延遲 t_p 可由振盪週期 T 與邏輯閘數量 n 算出，公式通常為 $t_p = T/2n$ 。

TTL 與 CMOS 的性能比較：

速度與電壓關係：40系列CMOS的傳遞延遲會隨著供應電壓VDD的提升而顯著縮短。TTL的切換速度較快，而CMOS則具有較寬的電源電壓工作範圍。

電源供應的重要性：實驗中必須嚴格區分訊號輸入與電源輸入，才能正常運作。

Experiment 1: Propagation Delay of 74-series Gates

1. Connect 5 **inverters** as shown in Fig. 1. **Don't forget to connect the power and ground from the power supply to any IC used in all of the experiments in this semester.** According to the datasheet, the IC is 74LS___. Its Vcc is pin ___ and GND is pin _____. **All 74 series ICs use a power of 5 Volts.**
2. Properly set the output of the FG to generate 5 Vpp (**0 V to 5 V**), 100 kHz pulses with a duty cycle of 25 %.
3. Use CH 1 and CH 2 in the scope to measure the average propagation delay of the five gates, and then calculate the average delay per gate. The delay per gate is $t_p = \text{_____}$.
4. Include a screen shot of the scope in your report.

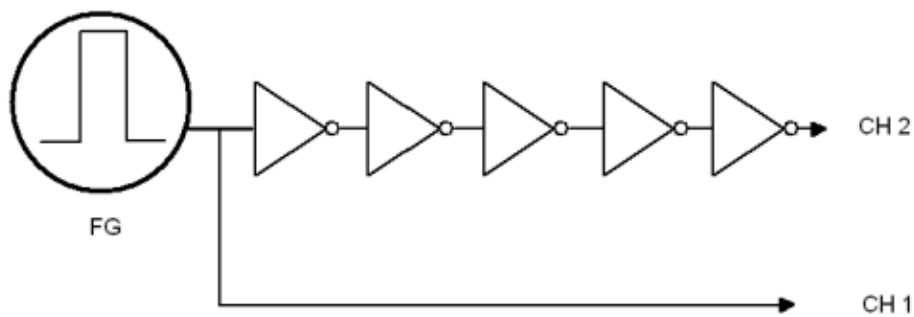
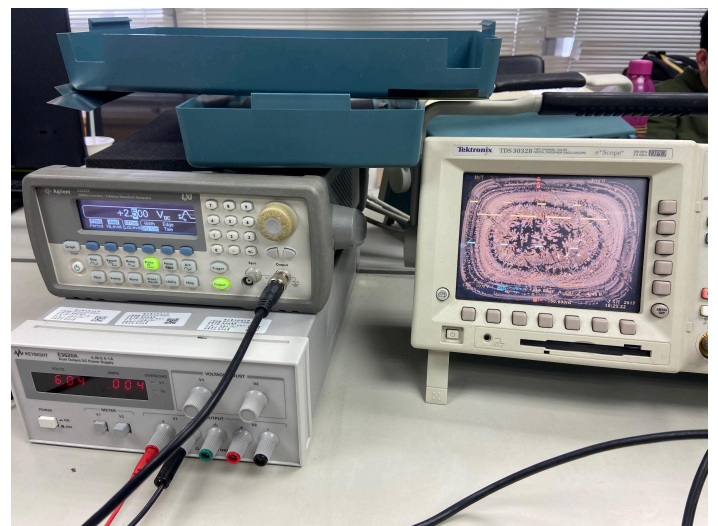
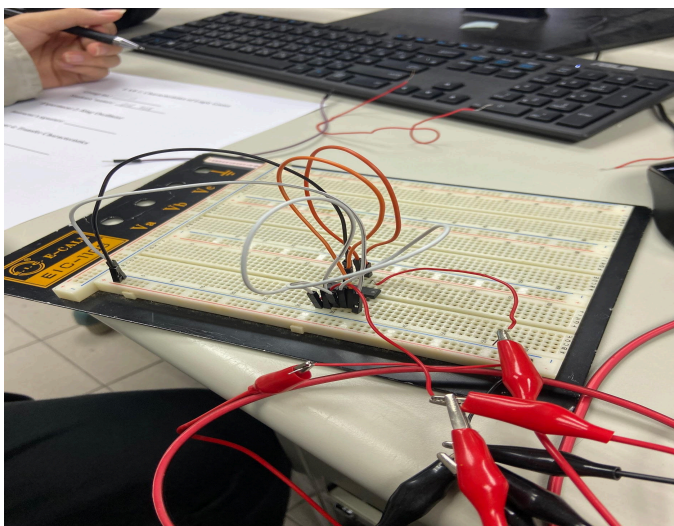


Fig. 1. Circuit to measure the propagation delay of a gate.



Experiment 2: Ring Oscillator (74 series)

1. Connect 5 inverters as shown in Fig. 2. No input is necessary. That is, the FG is not used here. **Remember the FG is not used to provide power to the ICs.**

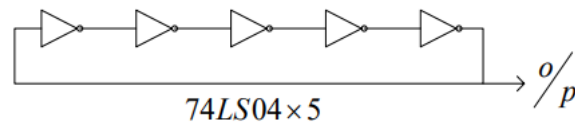
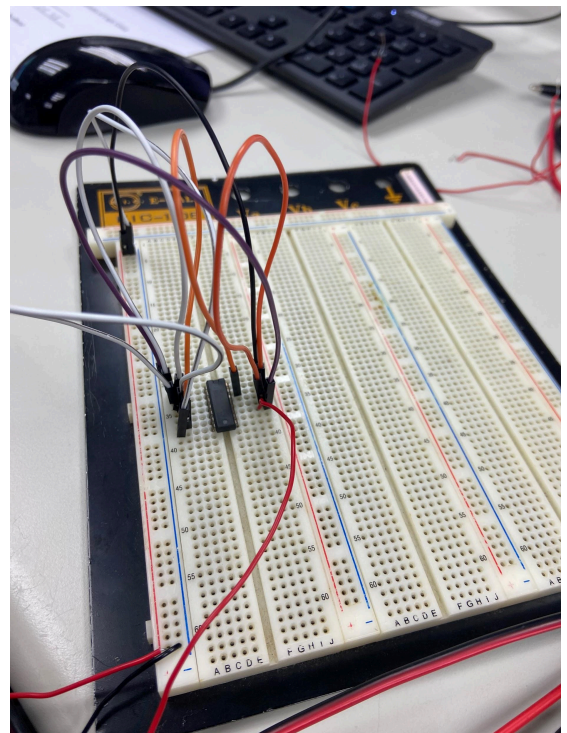
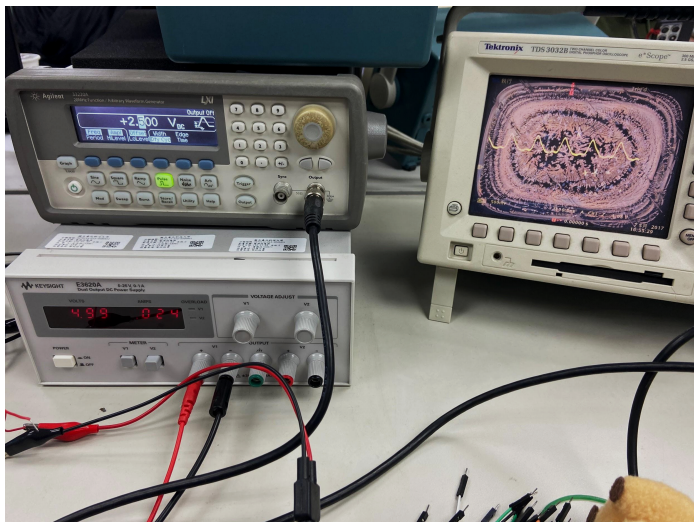


Fig. 2. Ring oscillator.

2. Observe the waveform at the output (**o/p**) using the scope and measure the period of the displayed waveform.
3. Ask the instructor to check your screen.
4. Based on the waveform, the average propagation delay for an inverter is t_p = _____ ns. **Explain how to calculate this value.**
5. Include a screen shot of the scope in your report.
6. Is the measured delay in this experiment consistent with that in Experiment 1? If not, why not?
7. If the sixth inverter is added to the circuit, what will happen? What will happen if the seventh is added? Explain why.
8. If your circuit has more than one IC, remember to give power to all ICs in the circuit.



Experiment 3: Ring Oscillator (40 series)

1. This experiment shows the differences between gates of the 74-series and those of 40-series.
2. Connect the circuit shown in Fig. 3.

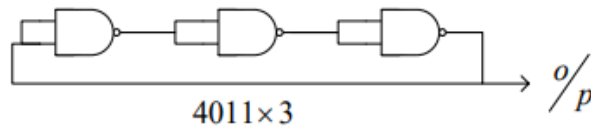
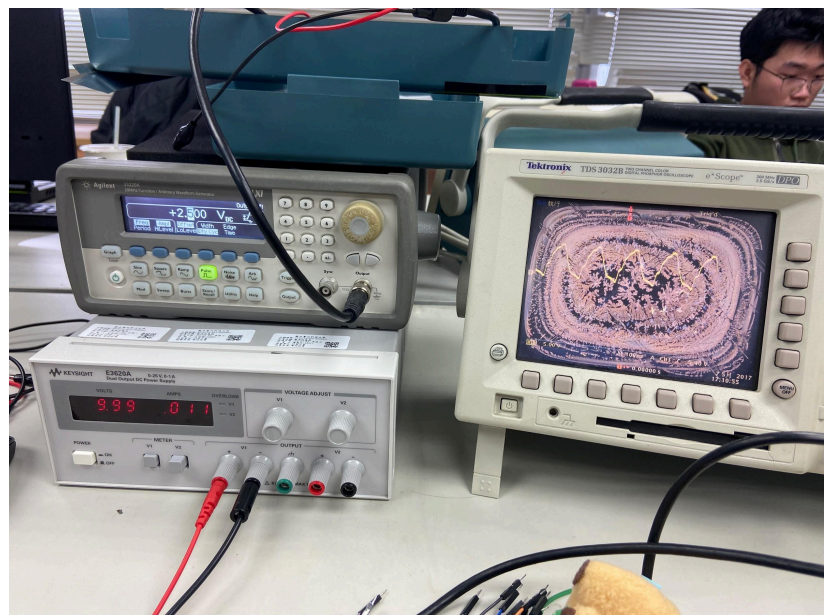
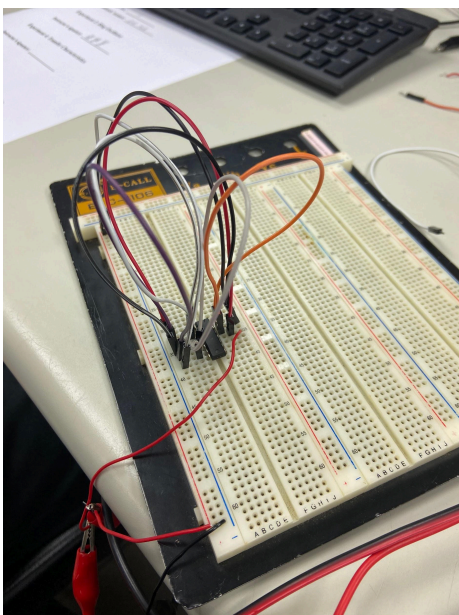


Fig. 3. Ring oscillator using 4000 series.

3. Set V_{DD} of the IC to +5V.
4. Calculate the propagation time for the gate. The delay time is ____ ns.
5. Set V_{DD} to +10V and +15V, respectively, and calculate the propagation time again. The delay time is ____ ns for 10 V and ____ ns for 15 V.
6. What is the relation between the propagation time and the supply voltage for a 40 series gate? Among the 74-series gates and 40-series gates, which series generally has a faster speed? Why?
7. Include a screen shot of the scope in your report.



Experiment 4: Transfer Characteristics (74 series)

1. Connect the circuit shown in Fig. 4. Use 74LS04 as the NOT gate.
2. Set the FG to generate $5V_{pp}$ (**0 V to 5 V**) triangular wave as the input to the gate.
3. Use the X-Y mode to plot the transfer characteristics of the gate.
4. **Adjust the oscilloscope knobs so that the origin of the transfer curve is on the center of the screen.**
5. Based on the transfer characteristics, when the input voltage is ____ V, the output of the gate changes its state (**midpoint** from high to low).
6. Ask the instructor to check your screen.
7. Include a screen shot of the scope in your report.

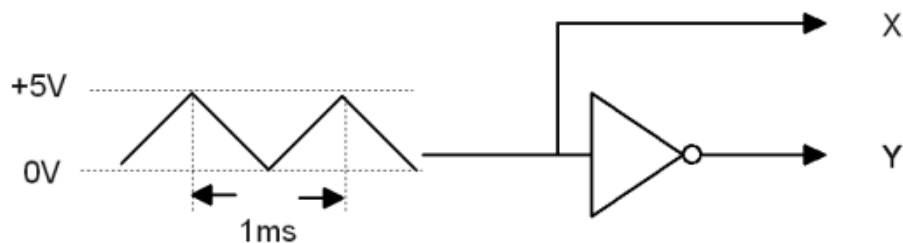


Fig. 4. Circuit to measure the transfer characteristics of a TTL gate.

